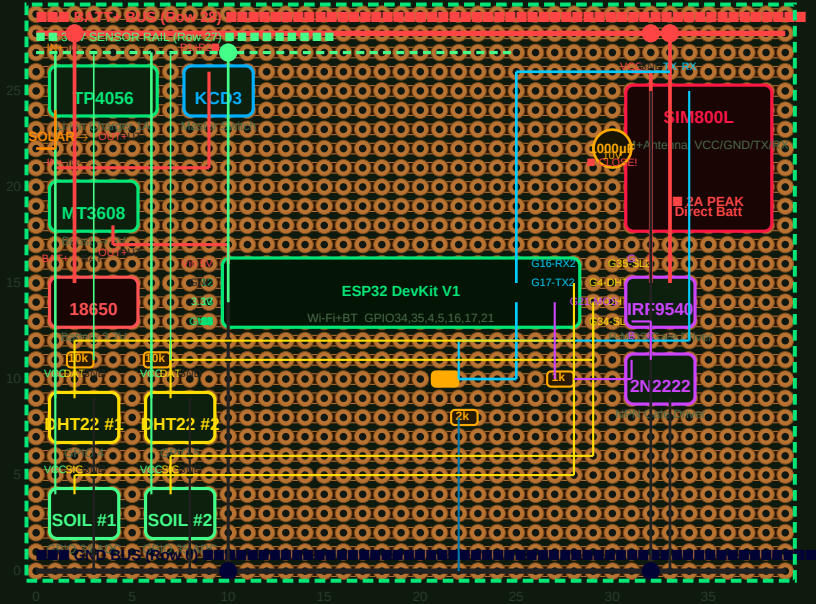


AGRISHIELD – Zero PCB Layout (40×30 holes, 0.1" pitch)

Print at 100% scale · A4 Landscape · Page 1 of 2: Component Placement & Copper Tracks

WIRE COLOUR LEGEND		
	Red	– Power / Battery+
	Black	– Ground (GND)
	Cyan	– UART (SIM800L)
	Yellow	– Sensor Signals
	Purple	– MOSFET Control
	Green	– 3.3V Sensor Rail
	Orange	– Solar Input
	Amber	– Resistors



10 × 0.1" pitch = 25.4mm (verify scale before drilling!)

AGRISHIELD – Assembly Guide & Wiring Reference

Page 2 of 2: Pin-to-Hole Reference, Build Sequence & Critical Warnings

1. COMPONENT HOLE POSITIONS (Col, Row)

ESP32 DevKit V1	Col 10–28, Row 13–16	(19×4)
TP4056 Charger	Col 1–6, Row 24–26	(6×3)
MT3608 Boost	Col 1–5, Row 18–20	(5×3)
KCD3 Switch (3-pin)	Col 8–11, Row 24–26	(4×3)
18650 Battery conn.	Col 1–5, Row 13–15	(5×3)
SIM800L Module	Col 31–38, Row 18–25	(8×8)
IRF9540 MOSFET	Col 31–34, Row 13–15	(4×3)
2N2222 NPN	Col 31–34, Row 9–11	(4×3)
DHT22 Sensor #1	Col 1–4, Row 7–9	(4×3)
DHT22 Sensor #2	Col 6–9, Row 7–9	(4×3)
Soil Sensor #1	Col 1–4, Row 2–4	(4×3)
Soil Sensor #2	Col 6–9, Row 2–4	(4×3)
1kΩ (base R)	Col 27–28, Row 10	(across 2 holes)
1kΩ (divider R1)	Col 21–22, Row 10	(across 2 holes)
2kΩ (divider R2)	Col 22, Row 8–10	(vertical)
10kΩ pull-up #1	Col 2–3, Row 11	(across 2 holes)
10kΩ pull-up #2	Col 6–7, Row 11	(across 2 holes)
1000µF 10V Cap	Col 29–30, Row 21–23	(radial, solder legs)

2. BUS RAILS (solder bridges)

GND Bus	Row 0, all 40 cols — solder all pads on this strip
BATT+ Bus	Row 28, all 40 cols — solder all pads on this strip
3.3V Rail	Row 27, col 0–25 — partial, sensor side only

3. STRIP CUTS (break copper track)

Use a sharp craft knife or stripboard cutter to cut these tracks:

✂ Row 5, between Col 14–15	→ Isolate SIM800L VCC from sensor side
✂ Row 10, between Col 8–9	→ Separate UART divider node
✂ Row 10, between Col 20–21	→ Separate UART divider node
✂ Row 15, between Col 6–7	→ Isolate sensor signal paths
✂ Row 15, between Col 25–26	→ Isolate UART pin tracks

4. BUILD SEQUENCE (do in order!)

- Test continuity of GND bus (row 0) with multimeter
- Solder GND bus row 0 – bridge all 40 pads
- Solder BATT+ bus row 28 – bridge all 40 pads
- Make all strip cuts (see Section 3) BEFORE inserting ICs
- Verify cuts with multimeter – no continuity across cut
- Insert & solder ESP32 DevKit V1 (headers or direct)
- Insert & solder TP4056 module
- Insert & solder MT3608 boost (pre-set to 5V before installing)
- Solder discrete resistors (1kΩ, 2kΩ, 10kΩ ×2)
- Insert & solder 2N2222 NPN (watch flat-side orientation)
- Insert & solder IRF9540 MOSFET TO-220 (heatsink if space)
- Solder 1000µF cap near SIM800L position (+ to BATT+)
- Install SIM800L last – test power section first
- Add sensor headers for DHT22 ×2 and Soil ×2
- Install KCD3 switch & 18650 battery connector last
- Apply conformal coating (MG Chem 422B) before final install

5. CRITICAL WARNINGS

- SIM800L must be powered from BATT+ (3.7–4.2V) via MOSFET — NEVER from ESP32 5V pin
- 1000µF capacitor MUST be within 2cm of SIM800L VCC/GND or module will crash on TX
- Voltage divider on ESP32 TX2→SIM800L RX is MANDATORY — 3.3V direct will damage SIM800L
- 2N2222 flat face orientation: flat side to flat side as printed on datasheet
- IRF9540 is P-Channel: Gate pulled LOW = ON, Gate pulled HIGH = OFF (inverted logic!)
- Use 22AWG or heavier wire for BATT+/MOSFET Drain/SIM800L VCC (2A peak current path)
- Pre-set MT3608 to exactly 5.0V with a multimeter BEFORE connecting ESP32
- All GND connections MUST share common row 0 bus — floating GND = data corruption
- Conformal coat ALL modules for Akure humidity — silicone or acrylic spray
- Verify ESP32 ADC pins 34 & 35 are INPUT ONLY — never drive them as outputs

6. VOLTAGE DIVIDER CALC (UART TX line)

ESP32 TX2 (GPIO 17) = 3.3V logic output

SIM800L RX max input = 2.8V

Divider: R1=1kΩ (series) + R2=2kΩ (shunt to GND)

$V_{out} = 3.3V \times 2k\Omega / (1k\Omega + 2k\Omega) = 3.3 \times 0.667 = 2.2V$ ✓

2.2V < 2.8V max — SIM800L RX is safe